

Multi-Modal Representation for Spatially Resolved Transcriptomics Based on Global Correlation and Dynamic Cluster Discovery

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Abstract

Constructing effective representations of spatial resolved transcriptomics (SRT) data, by appropriately characterizing the coherence in gene expression and histology with the spatial information of each sequencing spot, plays an important role in understanding the organization and function of complex tissues. Although much progress has been made, existing SRT representation methods typically establish local associative relationships for each spot only with those located in its surrounding spatial areas, thus failing to capture long-range correlations between distant regions. In addition, the absence of supervision signals on which cluster (with similar biological functions, pathological states or cell types) each spot should belong to also poses a great challenge in deriving an effective representation of SRT data. To this end, we propose a novel Multi-Modal SRT Representation Learning (MMSRL) method based on global spot correlation and dynamic cluster discovery. Specifically, given the gene expression and histological image, MMSRL first builds individual graph convolutional networks (GCNs) for these two modalities and bridges them through the adjacency matrix generated from the spatial locations of different spots. The extracted GCN features are then processed by a correlative self-attention operation to enhance their long-range correlations within each modality. Meanwhile, we also design a multi-modal interaction module (MMIM) to make these two-modal features interact with each other, and align their global correlation information across modalities. After that, we develop an attention-weighted fusion module (AWFM) to adaptively fuse the enhanced intra- and inter-modal features obtained above, so as to effectively integrate the multi-modal information. Finally, a dynamic cluster discovery process, which unsupervisedly assigns cluster labels to each spot, is incorporated to further refine the fused multi-modal representations in a contrastive learning manner.

CCS Concepts

• **Applied computing** → **Life and medical sciences**.

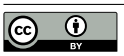
Keywords

Spatially resolved transcriptomics; Multi-modal representation

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1 Introduction

Spatially resolved transcriptomics (SRT) technology [5, 6, 21] employs advanced high-throughput sequencing and histological imaging techniques to map the gene expression profiles to their corresponding spatial locations (called spots) in the tissue slices. It allows us to explore the biological functions or pathological states of complex tissues in a spatial context, and thus opens new avenues for understanding the underlying mechanisms of biology systems [1, 22]. In order to fully utilize these SRT data in different practical applications, one of the most central steps is to construct effective representations for them, so that the gene expression and histological features, as well as the spatial location information, of each sequencing spot can be appropriately characterized.

For achieving this purpose, most existing methods [8, 24, 37, 42, 43] generate SRT representations solely from the gene expression data. They usually construct a K-nearest neighbor (KNN) graph based on the distances between different spot locations, and then build graph neural networks (GNNs) [41] from it to integrate the spatial structural information into the final gene expression representations. However, a major problem with these single-modal approaches is that they overlook the histological images which are often simultaneously acquired with gene expressions. These images not only provide high-resolution cellular morphological information that is complementary to sparse quantitative gene expression profiles [33], but also exhibit intrinsic correlations with them [38], together revealing the spatial heterogeneity of cells at different sequencing spots. Therefore, failure to leverage the complementarity and coherence between these two modalities will result in performance degradation of the above approaches.

To alleviate this problem, several new representation learning algorithms [2, 14, 17, 18, 31, 36] have been developed to deal with the multi-modal information of SRT data. According to how they utilize the histological images, these approaches can be roughly grouped into two categories: (1) Graph enhancement based methods [14, 17, 31, 36] typically calculate morphological similarity from the extracted image features and employ it to redefine the neighborhood relationships of each spot or augment the spatial adjacency matrix used for constructing GNNs; whereas (2) Multi-modal fusion based methods [2, 18] enable interactions between gene expression and associated histological features, directly learning joint representations by combining the complementary information across these two modalities. Despite obtaining impressive results, all the methods mentioned above still face two main challenges in constructing effective SRT representations:

The first one is that most complex tissue structures generally exhibit strong connections between different cells distributed in distant spatial regions [3, 28, 39]. However, previous studies that establish the associative relationships only within local areas around each sequencing spot (no matter whether they are enhanced by histological features or not) are unable to correctly capture such long-range correlation information, thus leading to decreased performance of downstream tasks. To show this, in Figure 1 (b) and (c), we visualize the related spots identified by SEDR [37] and SpaDAC [18], whose representations have a high similarity to that of an anchor spot (indicated by a red star). As can be seen, by only modeling the local spatial structures through a KNN graph, both of these

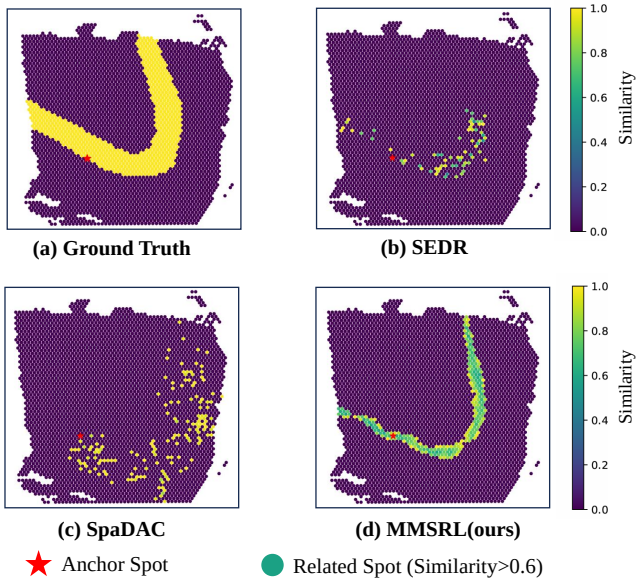


Figure 1: Visualization comparisons of identifying related spots that are highly similar to an random anchor in the human dorsolateral prefrontal cortex dataset, where similarity is measured by the normalized cosine distance. (a) shows the spots having the same cell type as the anchor spot. (b) and (c) highlight the spots identified by SEDR and SpaDAC that only consider the local spatial structures. (d) presents the potential related spots detected by our MMSRL method.

methods fail to accurately detect the related spots that are located far away from the anchor. The second challenge lies in the fact that there is no ground-truth annotation in the SRT data about which cluster (with similar biological functions, pathological states or cell types) each spot should belong to, and even the total number of clusters is not known. The absence of these supervision signals requires us to uncover potential correlations between different spots in a fully unsupervised manner, but without using a static method that defines the cluster number in advance.

To tackle these challenges, in this paper, we propose a novel Multi-Modal SRT Representation Learning (MMSRL) method based on global spot correlation and dynamic cluster discovery. Specifically, MMSRL first constructs individual graph convolutional networks (GCNs) [19] for the gene expression and histological image modalities, based on a shared adjacency matrix generated from the spatial locations of different spots. For the extracted two-modal GCN features, we then develop a correlative self-attention operation to separately enhance their intra-modal long-range correlations, and devise a multi-modal interaction module (MMIM) to combine them into unified inter-modal features as well as align their global correlation information across modalities. After that, these enhanced intra- and inter-modal features are adaptively fused through an attention-weighted fusion module (AWFM) to integrate the multi-modal information. In this way, MMSRL can effectively capture and exploit the global correlations of both modalities to generate the improved representation for each spot. Benefiting from

this, MMSRL is able to more accurately identify the anchor-related spots in tissues, as shown in Figure 1 (d). Finally, we incorporate a dynamic cluster discovery process to assign the pseudo cluster label to each spot. This process automatically splits a cluster with large internal variances into sub-clusters and merges those with high similarity into a new one, thus dynamically determining the number of clusters. With the assigned pseudo labels, MMSRL further refines the fused multi-modal representations obtained above by optimizing a contrastive learning objective. Both qualitative and quantitative results demonstrate that the proposed MMSRL method generates effective representations for SRT data, which lead to state-of-the-art performance in several downstream tasks, including spatial domain identification, tumor heterogeneity analysis, and gene expression enhancement, outperforming other competing methods. Our major contributions can be summarized as follows:

- We propose a novel multi-modal representation learning approach for SRT data, called MMSRL. By utilizing the spatial location information of sequencing spots as a bridge, MMSRL connects gene expression profiles and histology images, enabling effective integration of cross-modal features.
- We design the correlative self-attention operation and the multi-modal interaction module to fully exploit the global correlation information within each modality as well as between different modalities. In addition, a dynamic cluster discovery process is introduced to enhance the final multi-modal SRT representations in a contrastive learning manner without leveraging any prior knowledge of the cluster structures.
- We conduct extensive experiments on downstream tasks of stratified architectural pattern detection, temporal SRT data analysis, tumor heterogeneity analysis, mouse brain anterior analysis, and gene expression enhancement to evaluate the effectiveness and superiority of our MMSRL method over existing studies.

2 Related Work

Early spatial transcriptomics data analysis relied on traditional clustering methods such as k-means, mclust [12], SC3 [20], and Seurat(Louvain) [16], which used only gene expression data and ignored important spatial context information. Consequently, the results often exhibited discontinuities and failed to accurately represent the true tissue structure. This limitation hindered the understanding and analysis of complex biological tissues [13, 40].

To address this issue, the GNN-based methods have seen significant growth in spatial transcriptomics data analysis. Since the graph structure can effectively capture the complex relationships between gene expression and spatial distribution [11, 23, 25, 30], GNN has become an indispensable tool in this field. For instance, SEDR [37] uses a variational graph autoencoder (VGAE) to integrate gene expression and spatial information to obtain a latent low-dimensional representation. SpaceFlow [32] leverages spatially regularized deep GNN to embed spatial information into gene expression, promoting the discovery of latent representations. STAGATE [8] employs a graph attention autoencoder to fuse these two types of data. Spatial-MGCN [34] and MAFN [43] utilize multi-view GCNs combined with attention mechanisms to fuse gene expression and spatial information from multiple perspectives, effectively enhancing the comprehensive representation of SRT data. GraphST [26] uses a GNN method based on self-supervised contrastive learning, it learns discriminative spot representations by minimizing the feature space distance between spatially adjacent spots. stAA [10] introduces an adversarial training process into VGAE, enabling the acquired spot feature embeddings to better resolve spatial domain information. However, these methods only explore the integration of gene expression and spatial information, neglecting the histological image features corresponding to gene expression characteristics. In fact, the true potential of SRT data lies in its multi-modal nature, which provides multi-level support for in-depth analysis of SRT.

Recently, some methods have focused on the integration of three different modalities: gene expression, histological images, and spatial locations. stLearn [31] constructs smoother gene expression by leveraging morphological features extracted from histological images. SpaGCN [17] is based on the GCN method, which redefines the spatial adjacency relationships of each spot through image features. Although SpaGCN integrates histological image features, it simplifies the image features to the average RGB values, which may overlook more advanced features such as texture. Additionally, it does not construct a unified spatial multi-modal joint representation. DeepST [36] constructs a spatially enhanced gene expression matrix using extracted histological image features and employs GCN model to generate latent representations of the enhanced SRT data. SpaDAC [18] uses the GCN to generate spatial feature embeddings for gene expression data and histological images, and then employs simple feature concatenation to obtain a multi-modal representation. GRAS4T [14] uses histological image information to generate an enhanced graph representation, preserving the image knowledge critical for analytical tasks. However, these methods fail to adequately consider the complex global spot similarity, which not only exists within individual modalities but also manifests as the consistency and complementarity of global spots across different spatial multi-modal features in SRT data.

3 The Proposed MMSRL Method

As mentioned in Section 1, our MMSRL method mainly consists of four steps: (1) graph convolutional networks based feature extraction, (2) global correlation guided feature enhancement and interaction, (3) adaptive multi-modal feature fusion, and (4) dynamic cluster discovery driven feature refinement. In the following subsections, we will present the details of these steps.

3.1 GCN Based Feature Extraction

The main purpose of our MMSRL method is to reveal the coherence between gene expressions and histology images, and utilize their complementarity to generate more effective multi-modal SRT representations. To this end, it builds two GCN encoders, which are bridged by the spatial location information of all the spots, to separately extract features from the data of these two modalities.

Specifically, for the input gene expression profiles, we filter them following a standard preprocessing procedure to retain highly variable genes and thus obtain a normalized matrix $G \in \mathbb{R}^{N \times d_G}$, in which each row $g_i \in \mathbb{R}^{d_G}$ $(i = 1, 2, ..., N)$ represents the gene expression information of a particular sequencing spot. N and d_G indicate the number of spots and selected genes, respectively. Inspired by the masking mechanism of BERT [7], we randomly sample

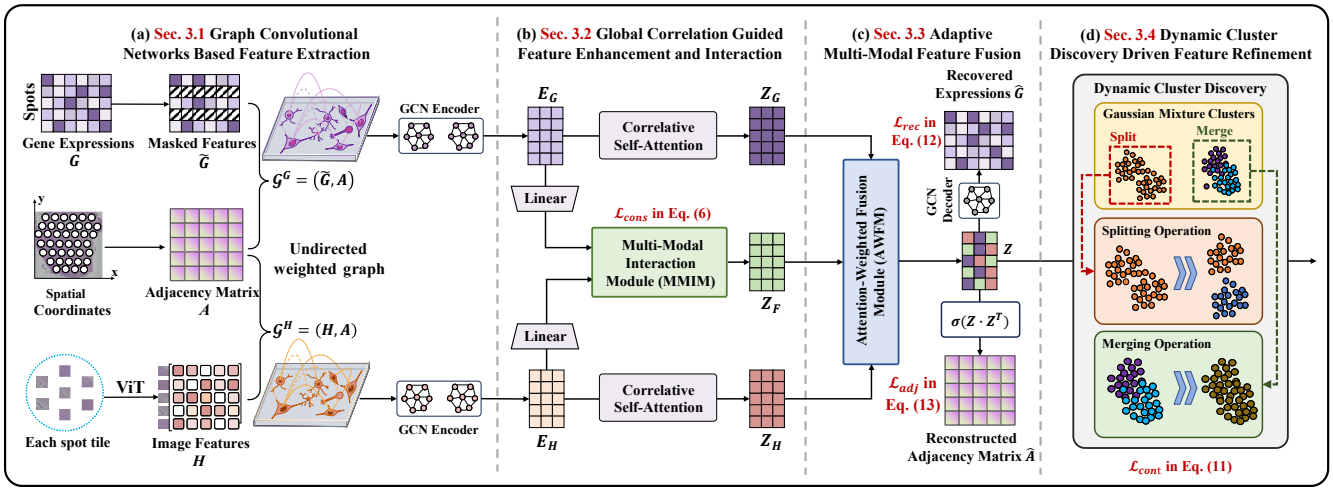


Figure 2: Overview of the proposed MMSRL method. There are four main steps: (a) Two graphs $\mathcal{G}_G = (\tilde{G}, A)$ and $\mathcal{G}_H = (H, A)$ are constructed for gene expressions and histology images, respectively, which are then fed into GCN encoders generate their relevant GCN features E_G and E_H (b) The correlative self-attention operation is applied to enhance the long-range correlations within E_G and E_H. At the same time, they align and interact with each other in the MMIM module. (c) The enhanced intra- and inter-modal features Z_G, Z_H and Z_F are adaptively fused in the AWFM module, resulting in the multi-modal representations Z. (d) The dynamic cluster discovery process is introduced to further refine Z in a constrastive learning manner.

a portion of spots and replace their corresponding rows in G with a learnable [MASK] vector, resulting in a masked gene expression matrix $\tilde{G} \in \mathbb{R}^{N \times d_G}$. During the training process, we will recover these masked rows by optimizing a reconstruction loss defined in Section 3.5. As for the histological image of a tissue slice, we first crop it into multiple tiles according to the spatial coordinates and sizes of different spots, and then extract morphological features from each of these spot tiles by adopting a pre-trained Vision Transformer (ViT) model [9]. The resulting feature matrix is denoted as $H \in \mathbb{R}^{N \times d_H}$, where d_H is the ViT output dimension.

Considering that $\tilde{G}$ and H are both associated with the same set of N sequencing spots, it is natural to construct two graphs $\mathcal{G}_G = (\tilde{G}, A)$ and $\mathcal{G}_H = (H, A)$ for them, respectively, based on an identical spot adjacency matrix $A \in \mathbb{R}^{N \times N}$, as illustrated in Figure 2 (a). Here, A serves as a bridge that connects the graphs of the two modalities. In order to correctly reflect the tissue structure, we calculate the entries in A depending on the spatial locations of spots through a Gaussian kernel function as follows:

$$A_{ij} = \exp(-\gamma d(i,j)^2), \ i \neq j, \tag{1}$$

where $d(i, j)$ denotes the Euclidean distance between the spatial coordinates of the i-th and j-th spots, γ is a positive factor controlling the scale of such distance. According to the above definition, A_{ij} will have a larger value for the closer located i-th and j-th spots, so the adjacency matrix A measures the strength of spatial connectivity between all the spots. This is different from the unweighted local KNN graphs commonly used in previous SRT studies.

After that, we separately feed the constructed gene expression graph $\mathcal{G}_G$ and histological image graph $\mathcal{G}_H$ into two GCN encoders to generate their corresponding GCN features as follows:

$$E_G = \mathcal{F}^G_{\text{GCN}}(\mathcal{G}_G), \quad E_H = \mathcal{F}^H_{\text{GCN}}(\mathcal{G}_H), \tag{2}$$

where $E_G, E_H \in \mathbb{R}^{N \times d}$ and d is their feature dimension. $\mathcal{F}_{\text{GCN}}(\cdot)$ represents a 2-layer GCN model, in which each convolution layer performs the following encoding operation:

$$E^{(i)} = \sigma_{\text{ReLU}}(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} E^{(i-1)} W^{(i)}). \tag{3}$$

Here, $E^{(i-1)}$ and $E^{(i)}$ denote the input and output node representations of the i-th layer, respectively. $W^{(i)}$ indicates the corresponding learnable parameters and $\sigma_{\text{ReLU}}(\cdot)$ is the ReLU activation function. $\tilde{A} = A + I$ adds self-loops to the original graph, with I being the identity matrix. $\tilde{D} = \text{diag}(D_1, D_2, ..., D_N)$ is a diagonal degree matrix where each entry on the diagonal is defined as the row-sum of $\tilde{A}$, satisfying $D_i = \sum_{j=1}^{N} \tilde{A}_{ij}$.

3.2 Global Correlation Guided Feature Enhancement and Interaction

To handle the challenge that spatially distant regions in tissues exhibit strong biological connections, our MMSRL method simultaneously enhances the long-range correlations within the GCN features E_G, E_H, and utilizes the global correlation information to facilitate the interaction between them, as shown in Figure 2 (b).

Within each single modality, we develop a correlative self-attention operation to process the encoded features E_G or E_H as:

$$Z_k = \text{Softmax}(\frac{E_k \cdot E_k^{\text{T}}}{\sqrt{d}}) E_k, \ k \in \{G, H\}. \tag{4}$$

Here, $E \cdot E^{\text{T}}$ calculates a similarity relationship matrix for all spots. Each row in this matrix represents the global dependencies between a particular spot and all other ones. $\sqrt{d}$ controls the similarity scale and the softmax function Softmax$(\cdot)$ computed along the column dimension normalizes the scaled similarity values into attentive weights within the range of $[0, 1]$. Based on this, every spot is able

to absorb useful information from the features of other spots with larger weights, even if they are located far away from each other. In this way, the long-range correlations between the cells distributed in spatially distant regions can be enhanced. It is worth noting that the definition in Eq. (4) removes the linear projectors in the original self-attention operation [15], because they may change the graph structural information embedded in the GCN features.

Across the two modalities, we devise a multi-modal interaction module (MMIM) that combines E_G and E_H, so as to generate the following fused inter-modal features:

$$Z_F = \mathcal{F}^F_{\text{Linear}}([Q_G \parallel Q_H]), \quad Q_G = E_G W_G,\ Q_H = E_H W_H, \tag{5}$$

where $W_G, W_H \in \mathbb{R}^{d \times d_C}$ are two projectors that map E_G and E_H into a d_C-dimensional common space, respectively. The operation $[\cdot \parallel \cdot]$ denotes the vector concatenation along the feature dimension, and $\mathcal{F}^F_{\text{Linear}}(\cdot)$ is a linear layer. By the process in Eq. (5), the informative features of E_G and E_H can be integrated into $Z_F \in \mathbb{R}^{N \times d}$ through Q_G and Q_H. In addition, to further exploit the underlying coherence between gene expressions and morphological features of histological images, we design the following loss function:

$$\begin{aligned} \mathcal{L}_{cons} &= \left\| \tilde{Q}_G \cdot \tilde{Q}_G^{\mathrm{T}} - \tilde{Q}_H \cdot \tilde{Q}_H^{\mathrm{T}} \right\|_F^2, \\ \tilde{Q}_G &= \mathcal{F}_{\text{Norm}}(Q_G), \quad \tilde{Q}_H = \mathcal{F}_{\text{Norm}}(Q_H). \end{aligned} \tag{6}$$

Here, $\|\cdot\|_F$ calculates the Frobenius norm of a matrix. $\tilde{Q}_G$ and $\tilde{Q}_H$ are obtained by applying the l_2 normalization $\mathcal{F}_{\text{Norm}}(\cdot)$ to the feature vector of each spot in Q_G and Q_H. According this definition, optimizing $\mathcal{L}_{cons}$ promotes the consistency of the global correlations $\tilde{Q} \cdot \tilde{Q}^{\mathrm{T}}$ between Q_G and Q_H, thereby aligning the tissue structure characterized by the two modality features. This can also guide Z_F in Eq. (5) to capture more cross-modal aligned information, which benefits the interaction between E_G and E_H.

3.3 Adaptive Multi-Modal Feature Fusion

After obtaining the intra-modal features Z_G and Z_H as well as the inter-modal features Z_F, our MMSRL method adaptively fuses them through an attention-weighted fusion module (AWFM) to fully exploit the complementarity between the two modalities [28] (see Figure 2 (c)). Specifically, considering that the features from different sources may have distinct impacts on the final fused representations, AWFM first generates the fusion weights for Z_G, Z_H and Z_F by defining the modal-attention operation as follows:

$$\Omega = \mathcal{F}_{\text{Norm}}(\text{Softmax}(\sigma_{\text{LeakyReLU}}([Z_G \parallel Z_H \parallel Z_F] W_\Omega))). \tag{7}$$

Here, $\sigma_{\text{LeakyReLU}}(\cdot)$ denotes the LeakyReLU activation function. The learnable parameter $W_\Omega \in \mathbb{R}^{3d \times 3}$ aggregates information from the three sets of features and maps $[Z_G \parallel Z_H \parallel Z_F]$ into a weight matrix $\Omega = [\omega_G \parallel \omega_H \parallel \omega_F] \in \mathbb{R}^{N \times 3}$. The three columns in Ω indicate the spot-wise attention weights of Z_G, Z_H and Z_F, respectively, and the l_2 normalization $\mathcal{F}_{\text{Norm}}(\cdot)$ is applied to further rescale their magnitudes. On this basis, AWFM performs the following adaptive fusion process for multi-modal features:

$$Z = \mathcal{F}^\Omega_{\text{Linear}}([\omega_G \odot Z_G \parallel \omega_H \odot Z_H \parallel \omega_F \odot Z_F]), \tag{8}$$

where $\odot$ is the broadcasting element-wise multiplication operation. In this process, ω_G, ω_H and ω_F describe the relative importance of different features specific to each spot, while the linear function $\mathcal{F}^\Omega_{\text{Linear}}(\cdot)$ characterizes the common fusion policy shared by all spots. By doing so, AWFM takes full advantage of both intra-modal and inter-modal information, thus enabling more effective feature fusion for generating the multi-modal representations $Z \in \mathbb{R}^{N \times d}$.

3.4 Dynamic Cluster Discovery

To deal with the second challenge that the ground-truth annotations of spot clusters are not available and even their total number is unknown, our MMSRL approach incorporates a dynamic cluster discovery process to pseudo-label each spot and accordingly refine the fused representations Z obtained in Eq. (8). Concretely, given an initial number of clusters C, we employ the Gaussian mixture model to divide N sequencing spots into C different clusters based on Z. Let $\mathcal{X}_c \subset Z$ represent the subset of spots assigned to the c-th cluster, $N_c = |\mathcal{X}_c|$ and $\mu_c \in \mathbb{R}^d$ denote the spot number and mean feature of this subset. To dynamically adjust the cluster number C, we define the splitting and merging operations below (Figure 2 (d)).

Splitting operation: To determine if the c-th cluster should be split, we first run the k-means algorithm with $k = 2$ on $\mathcal{X}_c$ to divide it into two sub-clusters $\mathcal{X}_{c,1}$ and $\mathcal{X}_{c,2}$, and then evaluate the following splitting criterion based on this division:

$$\mathcal{S}_c = \frac{\Gamma(N_{c,1}) L(\mathcal{X}_{c,1}; \theta) \Gamma(N_{c,2}) L(\mathcal{X}_{c,2}; \theta)}{\Gamma(N_c) L(\mathcal{X}_c; \theta)}. \tag{9}$$

Here, $N_{c,1} = |\mathcal{X}_{c,1}|$ and $N_{c,2} = |\mathcal{X}_{c,2}|$ denote the number of spots in the two sub-clusters. $\Gamma(\cdot)$ is the gamma function. $L(\cdot; \theta)$ is the marginal likelihood derived from a Normal-Inverse-Wishart prior distribution with θ as its hyperparameters. If $\mathcal{S}_c > 1$, it means that the marginal likelihood of these spots under the two sub-clusters is greater than their marginal likelihood under the original cluster. In this case, it is better to fit $\mathcal{X}_c$ with a two-component distribution. Therefore, we replace the original cluster with one of its sub-clusters and add the other one as a new cluster. The cluster centroids are also updated as $\mu_c \coloneqq \mu_{c,1}$, $\mu_{C+1} \coloneqq \mu_{c,2}$, where $\mu_{c,1}$ and $\mu_{c,2}$ are the mean vectors of the two sub-clusters, respectively.

Merging operation: Similarly, we evaluate whether to merge the i-th and j-th clusters through the criterion defined as follows:

$$\mathcal{M}_{ij} = \frac{\Gamma(N_i + N_j) L(\mathcal{X}_i \cup \mathcal{X}_j; \theta)}{\Gamma(N_i) L(\mathcal{X}_i; \theta) \Gamma(N_j) L(\mathcal{X}_j; \theta)}, \tag{10}$$

where $\cup$ is the union operation between two sets. When $\mathcal{M}_{ij} > 1$, it is better to utilize a single unified distribution to fit the spots in both $\mathcal{X}_i$ and $\mathcal{X}_j$. So we merge these two clusters into one and update the related cluster centroids as $\mu_i \coloneqq (\mu_i + \mu_j)/2$, $\mu_j \coloneqq \varnothing$.

After the above process, we can obtain C^* updated clusters and the corresponding cluster centroids μ_c $(c = 1, 2, ..., C^*)$. And for the i-th spot, it has been allocated into one of these clusters whose centroid is μ_i^*. With this, we refine its representation z_i (i.e., the i-th row in Z) by optimizing the following contrastive learning loss:

$$\mathcal{L}_{cont} = -\frac{1}{N} \sum_{i=1}^{N} \log \frac{\exp(z_i \cdot \mu_i^* / \tau)}{\sum_{c=1}^{C^*} \exp(z_i \cdot \mu_c / \tau)}, \tag{11}$$

where $\tau > 0$ is a temperature parameter. Benefiting from the dynamic cluster discovery process, $\mathcal{L}_{cont}$ can introduce additional supervision signals about the regions with similar biological functions, pathological states or cell types, even if they are not manually annotated. This enables our MMSRL model to generate interpretable

multi-modal representations that capture more semantic information for effective tissue structure analysis.

3.5 Training Loss of MMSRL

As mentioned in Section 3.1, we randomly mask the gene expression profiles of some spots (i.e., a portion of rows in $\tilde{G}$) with a learnable [MASK] vector. To ensure the multi-modal representations Z learned by our MMSRL method can correctly model the potential biological relationships between different spots and the coherence across different modalities, we attempt to recover these masked gene expression vectors from Z. To this end, let $\mathcal{K}$ denote the index set of masked spots, we formally define the recovery loss as follows:

$$\mathcal{L}_{rec} = \frac{1}{|\mathcal{K}|} \sum_{i \in \mathcal{K}} \|g_i - \hat{g}_i\|_2^2 . \tag{12}$$

Here, $\hat{g}_i$ is the i-th row of the recovered gene expression matrix $\hat{G}$, which is generated by decoding the graph $\mathcal{G}_Z = (Z, A)$ with a single-layer GCN model. Moreover, since MMSRL utilizes the adjacency matrix A as a bridge to connect the two modalities, we expect that the spatial structural information contained in A can be well embedded in the learned representations Z. Therefore, MMSRL also optimizes the following adjacency reconstruction loss:

$$\mathcal{L}_{adj} = \frac{1}{N^2} \left\| A - \hat{A} \right\|_F^2, \ \hat{A}_{ij} = \text{Sigmoid}\left(\frac{z_i \cdot z_j^{\mathrm{T}}}{\|z_i\|_2 \cdot \|z_j\|_2}\right), \ i \neq j, \tag{13}$$

where $\text{Sigmoid}(\cdot)$ represents the sigmoid function. Now, by linearly combining the loss functions defined in Eqs. (6), (11), (12), and (13), we obtain the total loss of our MMSRL method:

$$\mathcal{L}_{total} = \mathcal{L}_{cons} + \lambda_{cont}\mathcal{L}_{cont} + \lambda_{rec}\mathcal{L}_{rec} + \lambda_{adj}\mathcal{L}_{adj}, \tag{14}$$

where λ_{cont}, λ_{rec}, and λ_{adj} are three positive parameters that balance the relative importance of loss terms. We empirically set their values as $\lambda_{cont} = 1$, $\lambda_{rec} = 10$, and $\lambda_{adj} = 10$, respectively.

The overall workflow of MMSRL is summarized in Algorithm 1.

Algorithm 1 : MMSRL

Input: Multi-modal data, e.g., gene expression data G, histological image, and their shared spatial coordinates.
Parameter: The total epoch E, a initial C, temperature coefficient τ, and loss weight λ_{cont}, λ_{rec}, and λ_{adj}.

```
Extract image features H, initialize the adjacency matrix A in Eq. (1), L_cont = 0.
for e in 1, 2, ..., E do
    Construct graph G_G = (G̃, A) and graph G_H = (H, A) for the two modalities respectively.
    Generate GCN features E_G and E_H for two modalities in Eq. (2).
    Obtain intra-modality global correlation features Z_G and Z_H in Eq. (4).
    Obtain fused inter-modal features Z_F and global consistency alignment loss L_cons in Eq. (5) and Eq. (6).
    Fusing intra-modal and inter-modal features to a unified multi-modal representation Z in Eq. (7) and Eq. (8).
    if e ≥ 100 then
        if e mod 10 = 0 then
            Utilize the Gaussian Mixture Model to assign Z into C clusters and obtain cluster centers μ.
            Perform split and merge operations to update C* and μ* in Eq. (9) and Eq. (10).
            Update C = C*.
        end if
        Calculate dynamic prototype contrastive learning loss L_cont in Eq. (11).
    end if
    Calculate reconstruction loss L_rec and L_adj in Eq. (12) and Eq. (13).
    Train MMSRL by minimizing L_total in Eq. (14).
end for
Output Unified multi-modal representation Z.
```

4 Experiments

4.1 Experimental Settings

4.1.1 Datasets. Experiments are conducted on four publicly available spatial transcriptomics datasets, including the human dorsolateral prefrontal cortex (DLPFC) dataset [29], chicken heart dataset [27], the 10x Visium human breast cancer dataset [4] and the mouse brain anterior tissue dataset [43]. The detailed descriptions of these datasets and the data processing procedures are provided in Appendix A.1 and A.2, respectively.

4.1.2 Baselines. We select 9 state-of-the-art and representative approaches as baselines for performance comparison, including Spatial-MGCN [34], SEDR [37], GraphST [26], stAA [10], MAFN [43], DeepST [36], stLearn [31], SpaDAC [18], and GRAS4T [14]. More details of these methods are presented in Appendix A.3. All benchmark methods utilized in our experiments are configured with the default parameters as specified in their original papers.

4.1.3 Implementation Details. We adopt the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) as metrics for quantitative analysis. The detailed explanation of the metrics is provided in Appendix A.4. The introduction to the hyperparameters and the training process for MMSRL are given in Appendix A.5.

4.2 Quantitative Comparisons

We first conduct the quantitative analysis by comparing the spatial domain identification results (measured by ARI and NMI values) achieved by MMSRL and other benchmark methods on four datasets that covering different biological tissues.

As can be seen from Table 1, For DLPFC, in terms of the ARI metric, MMSRL achieves the best results on 10 slices while ranking second on the remaining ones. The leadership of MMSRL is especially significant on slices 151508, 151669 and 151670, outperforming other methods by at least 0.11, 0.13 and 0.20, respectively. As for the NMI metric, MMSRL again achieves the best performance on 9 slices, and only fails to place in the top three on slice 151675. The above observations also indicate that MMSRL have higher ARI and NMI median values with a relatively concentrated distribution. In contrast, the ARI and NMI values obtained by other methods are distributed more dispersed, indicating greater fluctuations in their performance. For the Chicken Heart dataset, MMSRL

Table 1: Comparison of ARI and NMI values between MMSRL and other methods on four datasets covering different biological tissues. The best and second-best results are highlighted in bold and underlined.

Methods	DLPFC												Chicken Heart				Breast	Mouse
	151507	151508	151509	151510	151669	151670	151671	151672	151673	151674	151675	151676	Day 4	Day 7	Day 10	Day 14	Cancer	Brain
ARI																		
Spatial-MGCN	0.68	0.51	0.71	0.61	0.56	0.48	0.62	0.76	0.57	0.50	0.46	0.53	0.35	0.39	0.44	0.46	0.60	0.44
SEDR	0.55	0.53	0.49	0.43	0.38	0.39	0.58	0.63	0.55	0.47	0.54	0.56	0.36	0.38	0.40	0.49	0.43	0.43
GraphST	0.48	0.49	0.52	0.50	0.48	0.46	0.61	0.63	0.60	0.43	0.55	**0.61**	0.34	0.39	0.39	0.44	0.53	0.46
stAA	0.59	0.47	0.46	0.42	0.47	0.49	0.62	0.60	0.55	0.50	0.53	0.57	0.37	0.40	0.46	0.48	0.58	0.43
MAFN	**0.68**	0.51	0.71	0.61	0.56	0.48	**0.82**	0.76	0.57	0.50	0.46	0.53	0.42	0.41	0.53	0.51	0.60	0.44
DeepST	0.52	0.39	0.46	0.43	0.34	0.32	0.62	0.53	0.50	0.47	0.53	0.50	0.39	0.37	0.45	0.44	0.53	0.28
stLearn	0.49	0.31	0.45	0.49	0.32	0.30	0.39	0.34	0.30	0.38	0.40	0.37	0.35	0.36	0.44	0.43	0.52	0.39
SpaDAC	0.58	0.52	0.68	0.71	0.51	0.49	0.68	0.72	0.61	0.51	0.52	0.54	0.40	0.40	0.49	0.48	0.54	0.41
GRAS4T	0.59	0.51	0.70	0.69	0.50	0.46	0.63	0.70	0.54	0.50	0.51	0.42	0.39	0.39	0.43	0.46	0.55	0.44
MMSRL	0.65	**0.64**	**0.77**	**0.75**	**0.69**	**0.69**	0.77	**0.79**	**0.63**	**0.54**	**0.58**	**0.61**	**0.62**	**0.56**	**0.69**	**0.63**	**0.66**	**0.51**
NMI																		
Spatial-MGCN	0.71	0.56	0.68	0.66	0.62	0.56	0.75	0.76	0.60	0.62	0.64	0.59	0.41	0.46	0.50	0.49	0.62	0.56
SEDR	0.70	0.66	0.63	0.64	0.57	0.53	0.69	0.71	0.68	0.60	**0.70**	0.63	0.40	0.41	0.44	0.52	0.63	0.72
GraphST	0.64	0.54	0.64	0.64	0.59	**0.68**	0.72	0.61	0.68	0.61	0.62	0.66	0.38	0.42	0.43	0.49	0.65	0.71
stAA	0.73	0.64	0.65	0.63	0.60	0.61	0.73	0.70	0.68	0.63	0.67	0.67	0.41	0.46	0.51	0.53	0.60	0.72
MAFN	**0.77**	0.51	0.72	0.68	0.63	0.60	**0.78**	0.73	0.67	0.62	0.60	0.68	0.44	0.54	0.59	0.56	0.68	0.68
DeepST	0.66	0.55	0.62	0.59	0.56	0.52	0.67	0.64	0.66	0.62	0.69	0.66	0.42	0.40	0.50	0.50	0.67	0.56
stLearn	0.64	0.53	0.62	0.64	0.49	0.44	0.54	0.47	0.49	0.54	0.55	0.54	0.39	0.41	0.53	0.53	0.60	0.71
SpaDAC	0.72	0.65	0.72	0.69	0.58	0.62	0.72	0.71	0.69	0.62	0.66	0.65	0.43	0.50	0.54	0.52	0.64	0.67
GRAS4T	0.69	0.64	0.71	0.71	0.62	0.54	0.69	0.71	0.64	0.62	0.65	0.56	0.43	0.51	0.53	0.51	0.64	0.64
MMSRL	0.75	**0.70**	**0.75**	**0.73**	**0.68**	**0.68**	0.74	**0.75**	**0.71**	**0.64**	0.67	**0.70**	**0.63**	**0.64**	**0.70**	**0.66**	**0.70**	**0.75**

exhibits clear advantages across different developmental stages. This indicates that MMSRL is more robust to tissue characteristic changes under dynamic time series, which allows it to provide reliable technical support for analyzing the complex spatial structural dynamics during the development of biological tissues (see Section 4.3.2). Moreover, MMSRL also achieves the best performance on the Breast Cancer and Mouse Brain datasets. All these results demonstrate the superiority and stability of MMSRL over existing methods. In addition, although DeepST, stLearn and GRAS4T incorporate histological image information, their performance is sometimes inferior to methods that do not use image data. This suggests the importance of effectively integrating the image information.

4.3 Downstream Task Qualitative Comparisons

4.3.1 Stratified Architectural Pattern Detection. In Figure 3, we visualize the stratified architectural pattern detection results obtained by different competing methods on DLPFC slice 151509. It can be observed that MMSRL identifies the cortical boundaries more accurately than other methods, making its results more similar to the manual annotations. Specifically, MMSRL is the only method that can correctly detect the thin strip-like stratified patterns for Layer 1 and Layer 2. And for Layer 5 and Layer 6 spots located in the lower-left corner of the slice, again only MMSRL is able to effectively distinguish them and detect their layered structure, while all other methods group them into the single layer. Moreover, the Uniform Manifold Approximation and Projection (UMAP) visualization results in Figure 3 (c) also show that the spots belonging to different layers are correctly clustered by MMSRL with clearer boundaries. This may be attributed to our designed correlative self-attention operation, which enhances the global correlations between different spots. Furthermore, we perform trajectory inference analysis using the PAGA algorithm based on the SRT representations learned by these methods. As shown in Figure 3 (d), MMSRL clearly reveal the trajectories between adjacent layers from White Matter (WM) to Layer 6 and then to Layer 1, whereas other methods exhibit varying degrees of confusion (see correlations between non-adjacent layers indicated by gray edges). These results demonstrate that the multi-modal representations learned by MMSRL improve the ability to reveal the underlying biological characteristics in SRT data.

4.3.2 Temporal SRT Data Analysis. Analyzing SRT data under time series can reveal the developmental patterns of tissues and organs. This further requires the SRT representation to effectively capture the spatio-temporal dynamic features. We employ the Chicken Heart dataset containing 12 slices collected at four key developmental stages to evaluate the effectiveness of our MMSRL method. Figure 4 compares the temporal analysis results of MMSRL with those of MAFN, which the best-performing benchmark according to Table 1. It can be seen that, MMSRL generates better results than MAFN that are more consistent with manual annotations. Concretely, MMSRL can correctly identify most spots of left ventricle (LV), right ventricle and atria across all stages. This means that MMSRL has the ability to produce similar multi-modal representations for the same cell type, thus enabling it to capture biological associations between different developmental stages and model the process of cell differentiation. Another important observation is that MMSRL can detect the epicardium spots around the outside of the chicken heart, while MAFN completely loses this information. This is also attributed to our correlative self-attention operation, allowing MMSRL to capture the long-range dependencies between those epicardium spots. Therefore, this result further validates the necessity of capturing the global correlations for SRT data representation.

MMSRL employs the masking mechanism during training and reconstructs gene expression in a self-supervised manner, exhibiting denoising capabilities. Therefore, MMSRL can also be used to efficiently filter out the sequencing noise, thus facilitating the accurate recognition of gene expression patterns. Please refer to Appendix B for gene expression enhancement.

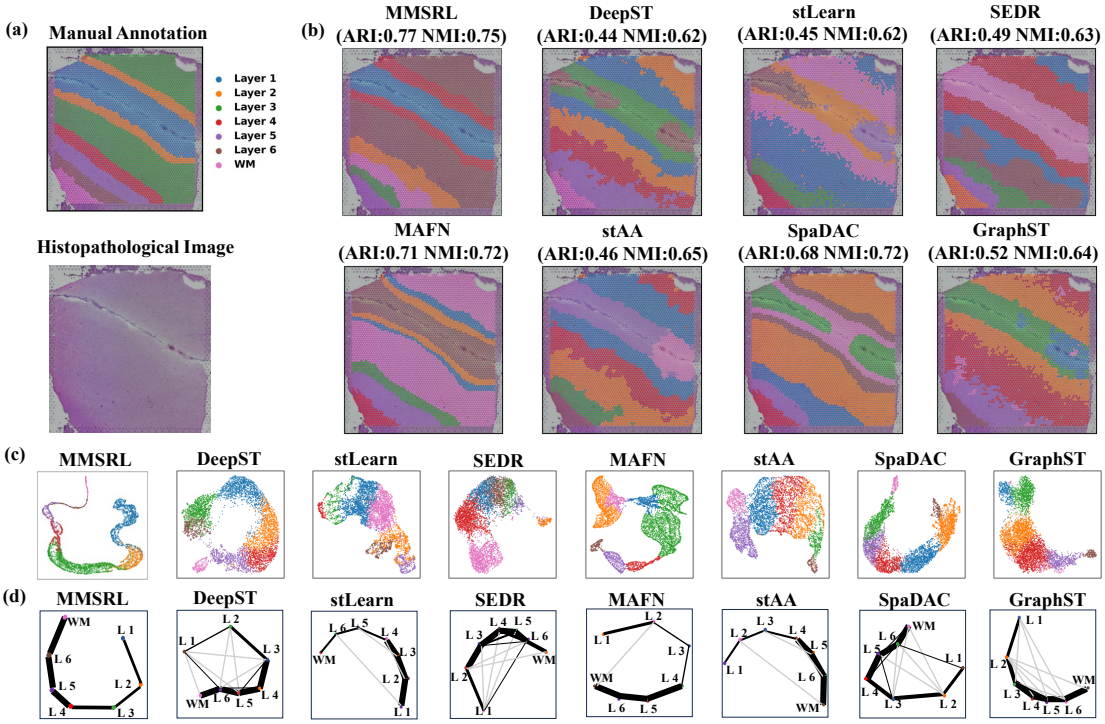


Figure 3: Performance comparisons of different methods on DLPFC datasets. (a)Annotation and histopathological image of slice 151509. (b) The domain identification results of different methods. (c) UMAP results. (d) Trajectory inference results.

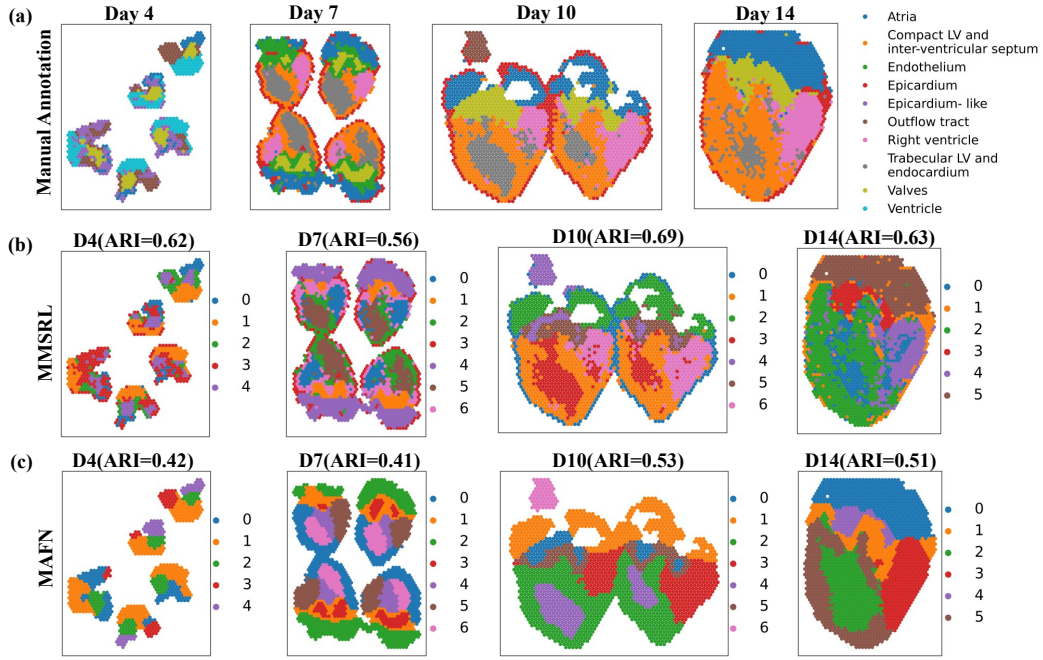


Figure 4: Analysis results on the Chicken Heart dataset. (a) The ground truth labels annotated in the original dataset. (b) The domains identified by MMSRL in chicken heart tissue. (c) Identification results of MAFN.

Table 2: Ablation study on the effect of each individual module in our method.

Self-attention	MMIM	AWFM	Dynamic Cluster	DLPFC 151509	Chicken Heart	Breast Cancer	Mouse Brain
×	×	×	×	0.38	0.33	0.36	0.25
✓	×	×	×	0.45	0.40	0.44	0.28
×	✓	×	×	0.48	0.43	0.46	0.30
×	×	✓	×	0.44	0.41	0.43	0.27
×	×	×	✓	0.49	0.45	0.47	0.32
✓	✓	×	×	0.52	0.47	0.49	0.35
✓	×	✓	×	0.50	0.45	0.48	0.33
✓	×	×	✓	0.53	0.49	0.51	0.36
×	✓	✓	×	0.55	0.51	0.52	0.38
×	✓	×	✓	0.58	0.56	0.56	0.42
×	×	✓	✓	0.57	0.54	0.54	0.39
✓	✓	✓	×	0.69	0.60	0.57	0.43
✓	✓	×	✓	0.70	0.63	0.60	0.45
✓	×	✓	✓	0.67	0.59	0.58	0.43
×	✓	✓	✓	0.71	0.61	0.58	0.46
✓	✓	✓	✓	**0.77**	**0.69**	**0.66**	**0.51**

4.4 Ablation Studies

4.4.1 Effect of Each Individual Module. Our proposed MMSRL framework consists of four major modules. To study their effects on the learned multi-modal representations, we construct several ablated models by gradually removing these modules from the full MMSRL framework, and then evaluate their ARI performance on the four datasets (the Day 10 slice is used for the evaluation on the Chicken Heart dataset). From the results shown in Table 2, we can find that: (1) When all the modules are removed, we obtain a baseline model that yields the worst results (lower than 0.30) across all datasets. (2) By integrating any individual module, the resulting ablated models improves the baseline model by at least 0.06, 0.07, 0.07, 0.02 and up to 0.11, 0.12, 0.11, 0.07 on the four datasets, respectively. (3) Such improvements can be more pronounced by further introduing more modules into the model. And eventually, the full MMSRL framework with all four modules consistently achieves the best results (0.77, 0.69, 0.66 and 0.51). The above observations demonstrate that each of the proposed modules has its own contribution to the final ARI results, and that combining more modules can promote facilitate cooperation among them, leading to more performance improvements.

4.4.2 Effect of Different Hyperparameter Settings. According to the loss function defined in Eq. (14), there are three parameters $\lambda_{cont}, \lambda_{rec}, \lambda_{adj}$ controlling the relative importance of different loss terms. We investigate their effects by varying their values, and the corresponding ARI results are reported in Table 3. By comparing the results in the first and fourth rows with those in the second and third rows, we can see that our MMSRL model performs relatively better when λ_{rec} and λ_{adj} have the same value. The recovery loss $\mathcal{L}_{rec}$ aims to recover the masked gene expressions, while the adjacency reconstruction loss $\mathcal{L}_{adj}$ focuses on restoring the global structure of the adjacency matrix. Achieving balance between these two losses can integrate both local gene expression feature information and global adjacency structural information into the SRT representations, thereby leading to improved performance. When further increasing λ_{rec} and λ_{adj} to 15, our model will pay too much attention to optimizing $\mathcal{L}_{rec}$ and $\mathcal{L}_{adj}$, and the importance of other two losses becomes relatively small, which leads to performance degradation. In addition, when λ_{cont} is greater than 1, the ARI values of our model drop significantly (from 0.77, 0.69, 0.66, 0.51 to 0.69, 0.60, 0.52, 0.46, respectively.) This may be because larger λ_{cont} can lead to solidification of the clustering structure, which may affect the effectiveness of unsupervised contrastive learning in the dynamic cluster discovery procedure.

Table 3: Ablation study on the effect of different hyperparameter settings.

λ_{cont}	λ_{rec}	λ_{adj}	DLPFC 151509	Chicken Heart	Breast Cancer	Mouse Brain
1	5	5	0.58	0.56	0.56	0.43
1	5	10	0.52	0.50	0.44	0.40
1	10	5	0.55	0.53	0.41	0.38
1	10	10	**0.77**	**0.69**	**0.66**	**0.51**
1	15	15	0.71	0.63	0.64	0.47
2	10	10	0.69	0.60	0.62	0.46
2	15	15	0.66	0.56	0.59	0.44

5 Conclusion

This study proposes a novel multi-modal representation learning method for SRT data, called MMSRL. To appropriately model the biological connections between spatially distant regions in tissues, MMSRL simultaneously enhances the long-range correlations of extracted two-modal GCN features, and facilitates the interactions between them by aligning their global correlation information. After that, the enhanced intra- and inter-modal features are adaptively fused to exploit the complementarity across the two modalities. Finally, to deal with the unsupervised nature of SRT data, MMSRL introduces the dynamic cluster discovery process to assign pseudo cluster labels to spots, and accordingly refine the learned representations with a contrastive learning loss. Extensive experimental validation demonstrates that MMSRL outperforms existing methods in several downstream tasks. The results underscore the potential of MMSRL to significantly advance our understanding of complex tissue organization and function, paving the way for more accurate and comprehensive spatial transcriptomics analysis.

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A Implementation Details

A.1 Datasets

All datasets used in this paper are publicly available data resources that have been released through relevant research publications or institutional platforms. Detailed information for each dataset is provided below.

Human dorsolateral prefrontal cortex: The human dorsolateral prefrontal cortex (DLPFC) dataset [29] consists of 12 tissue slices with detailed manual annotations depicting the structural distribution of six cortical layers and white matter (WM). Each slice contains 33,538 genes, with the number of spots ranging from 3,460 to 4,789, and includes 5 or 7 regions. The dataset can be downloaded from the link: http://spatial.libd.org/spatialLIBD/.

Chicken heart: The chicken heart dataset consists of 12 tissue slices, annotated across four key Hamburger-Hamilton ventricular developmental stages: Day 4 (5 slices), Day 7 (4 slices), Day 10 (2 slices), and Day 14 (1 slices). The dataset can be downloaded at: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE149457.

Human breast cancer: The 10x Visium human breast cancer dataset [4] contains 36,601 genes and 3,798 spots, with 20 manually annotated regions, primarily including healthy regions (Healthy), ductal carcinoma in situ/lobular carcinoma in situ (DCIS/LCIS), invasive ductal carcinoma (IDC), and tumor edges with low malignancy characteristics (Tumor edge). The dataset can be downloaded at: https://support.10xgenomics.com/spatial-gene-expression/datasets/.

Mouse anterior brain tissue: The mouse anterior brain tissue dataset [43] contains 32,285 genes and 2,695 spots, with 52 manually annotated regions. The dataset can be downloaded at: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSM3036911.

A.2 Data Processing

In our study, we first exclude spots situated outside the primary tissue areas for all datasets. Following this, we apply log transformation to the raw gene expression counts and perform library size normalization using the SCANPY package [35]. The normalized expression values are then scaled to achieve zero mean and unit variance. To streamline the analysis, we identify and utilize the top 3000 highly variable genes as the input for MMSRL.

For histological images, we first perform stain normalization on histopathological images to reduce the impact caused by uneven staining. Then, we extract the corresponding image for each spot based on its coordinates and size in the spatial transcriptomics data. Then, each spot image is divided into 4x4 patches, which are input into the pre-trained ViT model to extract the histological morphological features corresponding to each spot.

A.3 Baselines

To comprehensively demonstrate the advantages of the proposed MMSRL, we select several state-of-the-art and representative methods as baselines for comparison, including SEDR [37], GraphST [26], stAA [10], MAFN [43], DeepST [36], stLearn [31], SpaDAC [18], GRAS4T [14], SCANPY [35].

Spatial-MGCN is an SRT representation learning method based on multi-view GCN. It employs a multi-view GCN encoder to model the similarity between spatial structures and gene expressions.

SEDR uses a variational graph autoencoder (VGAE) to integrate gene expression and spatial information to obtain a latent low-dimensional representation.

GraphST uses a GNN-based method on self-supervised contrastive learning, it learns discriminative representations by minimizing the feature space distance between spatially adjacent spots.

stAA introduces an adversarial training process into VGAE, enabling the acquired spot feature embeddings to better resolve spatial domain information.

MAFN utilizes multi-view GCNs combined with attention mechanisms to fuse gene expression and spatial information from multiple perspectives, effectively enhancing the comprehensive representation of SRT data.

DeepST constructs a spatially enhanced gene expression matrix using extracted histological image features and employs GCN model to generate latent representations of the enhanced SRT data.

stLearn constructs smoother gene expression by leveraging morphological features extracted from histological images, effectively integrating tissue image information with gene expression data.

SpaDAC uses the GCN to generate spatial feature embeddings for gene expression and histological images, and then employs simple feature concatenation to obtain a multi-modal representation.

GRAS4T uses histological image information to generate an enhanced graph representation, preserving the image knowledge critical for analytical tasks.

SCANPY is a versatile and expandable framework tailored for the examination of gene expression data at the single-cell level, offering an all-encompassing approach for data analysis.

Among them, SEDR, GraphST, stAA, and MAFN are the latest GCN-based methods, each incorporating different techniques such as VGAE, self-supervised contrastive learning, adversarial training, and multi-view learning. In contrast, DeepST, stLearn, SpaDAC, and GRAS4T are the latest multi-modal learning methods that integrate gene expression, spatial location, and histological images to learn multi-modal representations of SRT data.

A.4 Evaluation Metrics

In this research, we employ the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) as evaluation metrics to evaluate the model's performance in spatial domain identification.

The ARI metric evaluates the consistency between the clustering labels predicted by the model and the actual ground truth labels:

$$ARI = \frac{RI - Ex[RI]}{\max(RI) - Ex[RI]}, \tag{15}$$

where $RI = \frac{2(TP+TN)}{N(N-1)}$, TP represents the number of true positive samples, TN is the number of true negative samples, N is the total number of samples. $Ex[RI]$ indicates the expected value that would be achieved by random labeling.

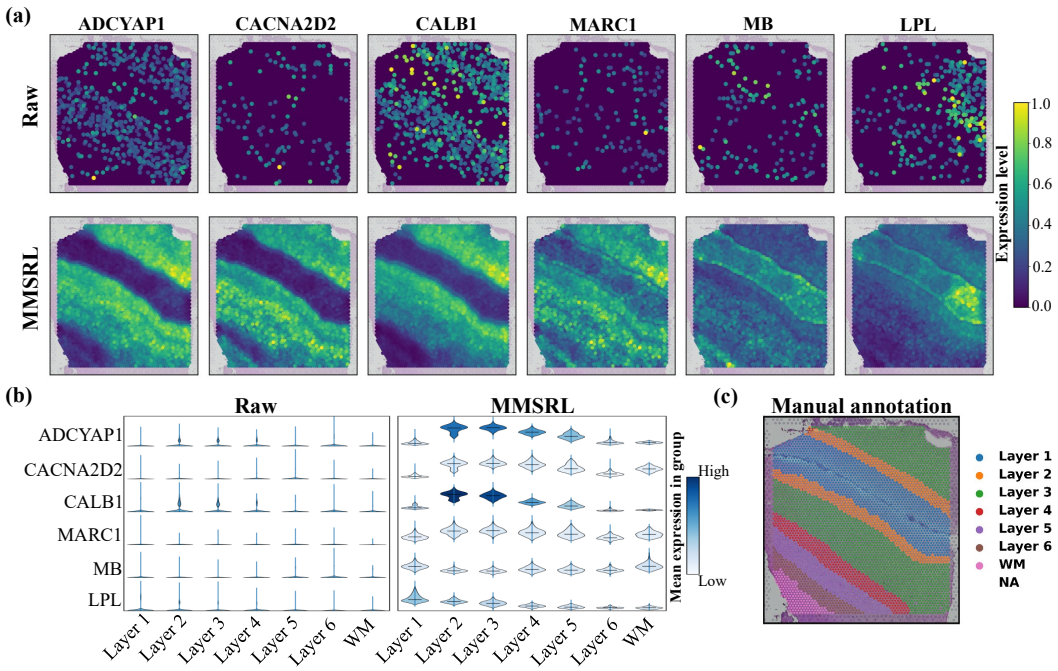


Figure 5: (a) The original and MMSRL-enhanced expression of layer-marker genes at 151509 DLPFC slice. (b) Violin plots of original and MMSRL-enhanced mean gene expression in different cortical layers. (c) Manual annotation of the 151509 slice.

The NMI metric is normalized mutual information, commonly used as an evaluation metric in unsupervised learning, particularly suitable for clustering tasks. The calculation form is as follows:

$$NMI(T, Y) = \frac{2MI(T, Y)}{H(T) + H(Y)}, \tag{16}$$

where $MI(\cdot, \cdot)$ represents the mutual information, $H(\cdot)$ denotes the information entropy.

A.5 Hyperparameters and Training Strategy

Certain models exhibit significant sensitivity to hyperparameter adjustments, with even small changes potentially causing substantial performance variations. Consequently, the effectiveness of these models often hinges on careful hyperparameter optimization. To minimize the influence of hyperparameters and highlight the robustness of MMSRL, we strive to keep the hyperparameters as consistent as we can across all datasets.

For all datasets, we standardize the raw gene expression counts using the SCANPY package and selected the top 3000 highly variable genes as the input for the MMSRL model. In all datasets, the gene expression data and histological images of each spot are finally embedded into 1*3000 vectors after the GCN encoding. In the DLPFC and chicken heart datasets, we set the initial number of clusters to 5 in the dynamic clustering discovery process, and to 20 for the other datasets. In all datasets, we set the value of temperature parameter in Eq. (11) as $\tau = 2$.

During the training process of the model, we use consistent training parameters for each dataset. We employ the Adam optimizer, set the learning rate to 0.0001, configured the weight decay to 5e-4, and stop the training process after 200 epochs to output the test results.

B Gene Expression Enhancement

In this section, we compare the original expression of six layer-marker genes (ADCYAP1, CACNA2D2, CALB1, MARC1, MB, and LPL) in the 151509 DLPFC slice with their MMSRL-recovered expression. As shown in Figure 5 (a) Raw, in the raw expression, these marker genes exhibit a strong confounding impact across different regions of the tissue, making it difficult to detect the layered patterns they should present. In contrast, the expressions enhanced by MMSRL exhibit much clearer spatial structures (see Figure 5 (a) MMSRL). For instance, CACNA2D2, as a marker gene of Layer 3, is not identified as a domain-specific gene in the raw expression data. In contrast, following MMSRL-based enhancement, it can be clearly recognized as a domain-specific gene of Layer 3. Furthermore, in the raw expression, genes such as ADCYAP1, CACNA2D2, and CALB1 exhibit sparse expression patterns, which blur the boundaries of spatial regions and make it difficult to discern distinct expression patterns. However, after MMSRL enhancement processing, the expression patterns of these genes in Layers 2 and 3 are significantly clearer, as can be clearly observed in Figure 5 (a) MMSRL. These results can also be observed in the violin plot of Figure 5 (b). In addition, the enrichment of MARC1 and MB within the WM region also becomes more pronounced (see Figure 5 (a) and (b)). These results indicate that MMSRL can integrate more useful information into the recovered gene expression, which may benefit from its multi-modal learning strategy that aggregates informative features from different modalities.